

Assignment 3
30 Marks
Deadline: Aug 18, 2026 (Tuesday)

Question - 1:

Why is a bias added to the actual exponent in the IEEE 754 floating-point representation, and how does this affect the encoding of both positive and negative exponents? [5]

Question - 2:

How does optimized multiplication improve efficiency and performance compared to traditional long multiplication, especially in terms of speed and computational complexity? [5]

Question - 3:

Multiply $(101101)_2$ with $(101)_2$ using optimized multiplier, you are required to show the whole simulation. [5]

Question - 4:

Perform the following multiplication using 24 bit IEEE-754 floating-point representation where the size of the fraction field is 14 bits. Check if the result has overflow or underflow. [5]

$$(-0.0101)_2 \times 2^{79} \text{ and } (10010.0101)_2 \times 2^{87}$$

Question - 5:

Convert the following code to RISC-V assembly language. [10]

```
float a = fp_array [0];
```

```
float b = fp_array [1];
```

```
int c = 0;
if (a < b) {
    c = (int) a;
} else {
    c = (int) b;
}
```

Where fp_array is a floating point word array base address of which is stored in x13, a is in f14 and b is in f7.

Note: Assume that the numbers stored in fp_array are single precision floating point numbers.